

T3 UV Lagrangian terms grouped by field content

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. The BSM-only UV expression is preferred; the full UV Lagrangian is used as a fallback when the BSM-only conversion is unavailable.

Field combinations 1 of 4

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	D	F	H	S_1
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1} \left(S_{2,j_1} \right)^\dagger D_{\mu_1} \left(S_{2,j_1} \right)$	$\begin{aligned} & -\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F \\ & +\overline{y_{2,i_1,r_1}} \overline{F_{j_1,r_1}} P_L L_{a_1,i_1} S_{2,j_1} \text{Bar}(\text{T3Y2CG})_{a_1,j_1,j_1} \\ & +\overline{L_{a_1,i_1}} P_R F_{j_1,r_1} y_{2,i_1,r_1} \mathcal{I}_{Y^2,a_1,j_1,j_1} S_{2,j_1}^\dagger \end{aligned}$	$\begin{aligned} & +\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H^2,a_1,a_2,j_1,j_2} {}^1\lambda_{H^2} {}^1 \\ & +\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H^2,a_1,a_2,j_2,j_1} {}^1\lambda_{H^2} {}^1 \\ & +\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H^2,a_1,a_2,j_1,j_2} {}^2\lambda_{H^2} {}^2 \\ & +\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H^2,a_1,a_2,j_2,j_1} {}^2\lambda_{H^2} {}^2 \end{aligned}$	$\begin{aligned} & -S_1 S_1^\dagger M_{S_1}^2 \\ & +S_1 S_{2,j_2} S_1^\dagger S_{2,j_1}^\dagger \mathcal{I}_{12,j_1,j_2} {}^1\lambda_{12} {}^1 \\ & +\frac{1}{2} S_1^2 \left(S_1^\dagger \right)^2 \lambda_{S_1} {}^1 \end{aligned}$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$\begin{aligned} & +D_{\mu_1} \left(S_{1,j_1} \right)^\dagger D_{\mu_1} \left(S_{1,j_1} \right) \\ & +D_{\mu_1} \left(S_{2,j_1} \right)^\dagger D_{\mu_1} \left(S_{2,j_1} \right) \end{aligned}$	$\begin{aligned} & -\overline{F_{r_1}} F_{r_1} M_F \\ & +\overline{L_{a_1,i_1}} C P_R \overline{F_{r_1}}^T S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{Y^1,a_1,j_1} \\ & +\overline{y_{2,i_1,r_1}} \overline{F_{r_1}} P_L L_{a_1,i_1} S_{2,j_1} \text{Bar}(\text{T3Y2CG})_{a_1,j_1} \\ & +\overline{y_{1,i_1,r_1}} \overline{L_{a_1,i_1}}^T C P_L F_{r_1} \text{Bar}(\text{T3Y1CG})_{a_1,j_1} S_{1,j_1}^\dagger \\ & +\overline{L_{a_1,i_1}} P_R F_{r_1} y_{2,i_1,r_1} \mathcal{I}_{Y^2,a_1,j_1} S_{2,j_1}^\dagger \end{aligned}$	$\begin{aligned} & +\overline{\lambda_{T3}} S_{2,j_2} \text{Bar}(\text{T3MixCG})_{a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger \\ & +H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{T^3,a_1,a_2,j_2,j_1} \lambda_{T^3} S_{2,j_1}^\dagger \\ & +H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H^1,a_1,a_2,j_1,j_2} {}^1\lambda_{H^1} {}^1 \\ & +H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H^1,a_1,a_2,j_1,j_2} {}^2\lambda_{H^1} {}^2 \\ & +H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H^2,a_1,a_2,j_1,j_2} {}^1\lambda_{H^2} {}^1 \\ & +H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H^2,a_1,a_2,j_1,j_2} {}^2\lambda_{H^2} {}^2 \end{aligned}$	
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$\begin{aligned} & +D_{\mu_1} \left(S_{1,j_1} \right)^\dagger D_{\mu_1} \left(S_{1,j_1} \right) \\ & +D_{\mu_1} \left(S_{2,j_1} \right)^\dagger D_{\mu_1} \left(S_{2,j_1} \right) \end{aligned}$	$\begin{aligned} & -\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F \\ & +\overline{L_{a_1,i_1}} C P_R \overline{F_{j_1,r_1}}^T S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{Y^1,a_1,j_1,j_1} \\ & +\overline{y_{2,i_1,r_1}} \overline{F_{j_1,r_1}} P_L L_{a_1,i_1} S_{2,j_1} \text{Bar}(\text{T3Y2CG})_{a_1,j_1,j_1} \\ & +\overline{y_{1,i_1,r_1}} \overline{L_{a_1,i_1}}^T C P_L F_{j_1,r_1} \text{Bar}(\text{T3Y1CG})_{a_1,j_1,j_1} S_{1,j_1}^\dagger \\ & +\overline{L_{a_1,i_1}} P_R F_{j_1,r_1} y_{2,i_1,r_1} \mathcal{I}_{Y^2,a_1,j_1,j_1} S_{2,j_1}^\dagger \end{aligned}$	$\begin{aligned} & +\overline{\lambda_{T3}} S_{2,j_2} \text{Bar}(\text{T3MixCG})_{a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger \\ & +H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{T^3,a_1,a_2,j_2,j_1} \lambda_{T^3} S_{2,j_1}^\dagger \\ & +H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H^1,a_1,a_2,j_1,j_2} {}^1\lambda_{H^1} {}^1 \\ & +H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H^1,a_1,a_2,j_1,j_2} {}^2\lambda_{H^1} {}^2 \\ & +H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H^2,a_1,a_2,j_1,j_2} {}^1\lambda_{H^2} {}^1 \\ & +H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H^2,a_1,a_2,j_1,j_2} {}^2\lambda_{H^2} {}^2 \end{aligned}$	
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+D_{\mu_1} \left(S_{1,j_1} \right)^\dagger D_{\mu_1} \left(S_{1,j_1} \right)$	$\begin{aligned} & -\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F \\ & +\overline{L_{a_1,i_1}} C P_R \overline{F_{j_1,r_1}}^T S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{Y^1,a_1,j_1,j_1} \\ & +\overline{y_{1,i_1,r_1}} \overline{L_{a_1,i_1}}^T C P_L F_{j_1,r_1} \text{Bar}(\text{T3Y1CG})_{a_1,j_1,j_1} S_{1,j_1}^\dagger \end{aligned}$	$\begin{aligned} & +\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H^1,a_1,a_2,j_1,j_2} {}^1\lambda_{H^1} {}^1 \\ & +\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H^1,a_1,a_2,j_2,j_1} {}^1\lambda_{H^1} {}^1 \\ & +\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H^1,a_1,a_2,j_1,j_2} {}^2\lambda_{H^1} {}^2 \\ & +\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H^1,a_1,a_2,j_2,j_1} {}^2\lambda_{H^1} {}^2 \end{aligned}$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$\begin{aligned} & +D_{\mu_1} \left(S_{1,j_1} \right)^\dagger D_{\mu_1} \left(S_{1,j_1} \right) \\ & +D_{\mu_1} \left(S_{2,j_1} \right)^\dagger D_{\mu_1} \left(S_{2,j_1} \right) \end{aligned}$	$\begin{aligned} & -\overline{F_{j_1,r_1}} F_{j_1,r_1} M_F \\ & +\overline{L_{a_1,i_1}} C P_R \overline{F_{j_1,r_1}}^T S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{Y^1,a_1,j_1,j_1} \\ & +\overline{y_{2,i_1,r_1}} \overline{F_{j_1,r_1}} P_L L_{a_1,i_1} S_{2,j_1} \text{Bar}(\text{T3Y2CG})_{a_1,j_1,j_1} \\ & +\overline{y_{1,i_1,r_1}} \overline{L_{a_1,i_1}}^T C P_L F_{j_1,r_1} \text{Bar}(\text{T3Y1CG})_{a_1,j_1,j_1} S_{1,j_1}^\dagger \\ & +\overline{L_{a_1,i_1}} P_R F_{j_1,r_1} y_{2,i_1,r_1} \mathcal{I}_{Y^2,a_1,j_1,j_1} S_{2,j_1}^\dagger \end{aligned}$	$\begin{aligned} & +\overline{\lambda_{T3}} S_{2,j_2} \text{Bar}(\text{T3MixCG})_{a_1,a_2,j_1,j_2} H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger \\ & +H_{a_1} H_{a_2} S_{1,j_2} \mathcal{I}_{T^3,a_1,a_2,j_2,j_1} \lambda_{T^3} S_{2,j_1}^\dagger \\ & +\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H^1,a_1,a_2,j_1,j_2} {}^1\lambda_{H^1} {}^1 \\ & +\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H^1,a_1,a_2,j_2,j_1} {}^1\lambda_{H^1} {}^1 \\ & +\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H^1,a_1,a_2,j_1,j_2} {}^2\lambda_{H^1} {}^2 \\ & +\frac{1}{2} H_{a_2} S_{1,j_2} H_{a_1}^\dagger S_{1,j_1}^\dagger \mathcal{I}_{H^1,a_1,a_2,j_2,j_1} {}^2\lambda_{H^1} {}^2 \\ & +\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H^2,a_1,a_2,j_1,j_2} {}^1\lambda_{H^2} {}^1 \\ & +\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H^2,a_1,a_2,j_2,j_1} {}^1\lambda_{H^2} {}^1 \\ & +\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H^2,a_1,a_2,j_1,j_2} {}^2\lambda_{H^2} {}^2 \\ & +\frac{1}{2} H_{a_2} S_{2,j_2} H_{a_1}^\dagger S_{2,j_1}^\dagger \mathcal{I}_{H^2,a_1,a_2,j_2,j_1} {}^2\lambda_{H^2} {}^2 \end{aligned}$	

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	constant	F	D	S_1	D
T3-A	0	1	0	3	1	2	$\frac{1}{2}$		$-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^1$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^2 \lambda_{S_2}^2$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu 1} D_{\mu 1} (F_{j_1,r_1})$		$+D_{\mu 1} (S_1)^\dagger D_{\mu 1} (S_1)$	
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^1$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^1$	$+i\overline{F_{r_1}} \gamma^{\mu 1} D_{\mu 1} (F_{r_1})$				
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^1$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^1$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu 1} D_{\mu 1} (F_{j_1,r_1})$				
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-M_{S_2}^2 S_2 S_2^\dagger$ $+S_2 S_{1,j_2} S_2^\dagger S_{1,j_1}^\dagger \mathcal{I}_{12,j_1,j_2}^1 \lambda_{12}^1$ $+\frac{1}{2} S_2^2 (S_2^\dagger)^2 \lambda_{S_2}^1$	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^1$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^2 \lambda_{S_1}^2$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu 1} D_{\mu 1} (F_{j_1,r_1})$			
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-S_{1,j_1} S_{1,j_1}^\dagger M_{S_1}^2$ $-M_{S_2}^2 S_{2,j_1} S_{2,j_1}^\dagger$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^1 \lambda_{12}^1$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^2 \lambda_{12}^2$ $+S_{1,j_3} S_{2,j_4} S_{1,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}^3 \lambda_{12}^3$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^1 \lambda_{S_1}^1$ $+\frac{1}{2} S_{1,j_3} S_{1,j_4} S_{1,j_1}^\dagger S_{1,j_2}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}^2 \lambda_{S_1}^2$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^1 \lambda_{S_2}^1$ $+\frac{1}{2} S_{2,j_3} S_{2,j_4} S_{2,j_1}^\dagger S_{2,j_2}^\dagger \mathcal{I}_{S_2,j_1,j_3,j_2,j_4}^2 \lambda_{S_2}^2$	$+i\overline{F_{j_1,r_1}} \gamma^{\mu 1} D_{\mu 1} (F_{j_1,r_1})$				

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1 F$	$S_1 H$	$S_2 D$	$S_2 F$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\overline{L_{a_1,i_1}} C P_R \overline{F_{j_1,r_1}}^T S_1 y_{1,i_1,r_1} \mathcal{I}_{y1,a_1,j_1}$ $+\overline{y_{1,i_1,r_1}} L_{a_1,i_1}^T C P_L F_{j_1,r_1} \text{Bar}(\text{T3Y1CG})_{a_1,j_1} S_1^\dagger$	$+\overline{\lambda_{\text{T3}}} S_{2,j_1} \text{Bar}(\text{T3MixCG})_{a_1,a_2,j_1} H_{a_1}^\dagger H_{a_2}^\dagger S_1^\dagger$ $+H_{a_1} H_{a_2} S_1 \mathcal{I}_{\text{T3},a_1,a_2,j_1} \lambda_{\text{T3}} S_{2,j_1}^\dagger$ $+H_{a_2} S_1 H_{a_1}^\dagger S_1^\dagger \mathcal{I}_{\text{H1},a_1,a_2}^1 \lambda_{\text{H1}}^1$		
T3-B	0	2	0	2	1	1	$\frac{1}{2}$				
T3-C	0	2	0	2	1	3	$\frac{1}{2}$				
T3-D	0	3	0	1	1	2	$\frac{1}{2}$			$+D_{\mu_1}(S_2)^\dagger D_{\mu_1}(S_2)$	$+\overline{y_{2,i_1,r_1}} \overline{F_{j_1,r_1}} P_L L_{a_1,i_1} S_2 \text{Bar}(\text{T3Y2CG})_{a_1,j_1}$ $+\overline{L_{a_1,i_1}} P_R F_{j_1,r_1} y_{2,i_1,r_1} \mathcal{I}_{y2,a_1,j_1} S_2^\dagger$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$				

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Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2	H
T3-A	0	1	0	3	1	2	$\frac{1}{2}$		
T3-B	0	2	0	2	1	1	$\frac{1}{2}$		
T3-C	0	2	0	2	1	3	$\frac{1}{2}$		
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_{T3}S_2^\dagger$ $+\overline{\lambda_{T3}}S_2\text{Bar}(\text{T3MixCG})_{a_1,a_2,j_1}H_{a_1}^\dagger H_{a_2}^\dagger S_{1,j_1}^\dagger$ $+H_{a_2}S_2H_{a_1}^\dagger S_2^\dagger\mathcal{I}_{H2,a_1,a_2}^1\lambda_{H2}^1$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$		